



Simulations of *NMR* Relaxation in a Real Porous Structure: Pre-asymptotic Behavior to the Localization Regime

J. L. Gonzalez¹ · E. L. de Faria² · Marcelo P. Albuquerque² ·
Marcio P. Albuquerque² · Clécio R. Bom³ · J. C. C. Freitas¹ · Maury D. Correia⁴

Received: 5 October 2019 / Revised: 13 April 2020 / Published online: 5 May 2020
© Springer-Verlag GmbH Austria, part of Springer Nature 2020

Abstract

The knowledge of the different physical mechanisms determining the relaxation of the magnetization is crucial to obtain reliable petrophysical parameters from *NMR* experiments. Some studies have focused on asymptotic mechanisms providing simple expressions for the relaxation rate in terms of parameters of the system. The main difficulties in solving this problem are related to the existence of intrinsic magnetic inhomogeneities, in addition to the complex structure of the porous geometry found in real systems. In this work, simulations of the relaxation of the transverse magnetization were performed on a real pore structure obtained from X-ray microtomography images. Also, a statistical method was used to introduce a local magnetic non-homogeneity, resembling local field distributions discussed in several works in the literature. The conditions of our simulations allowed the study of relaxation processes mainly influenced by local magnetic gradients, while surface relaxation became irrelevant. The dependence of the relaxation rate on the inter-echo spacing was studied, where for low spacings, the results reproduced those expected in the free-diffusion limit. On the other hand, for high inter-echo spacings, the relaxation rate deviates from the expected dependence, indicating the emergence of a new relaxation mechanism. This high-spacing behavior was associated with a pre-asymptotic localization regime. Our data suggest a logarithmic dependence of the relaxation rate with the inter-echo spacing in this transition regime.

✉ J. L. Gonzalez
jlga.1966@gmail.com

¹ Universidade Federal do Espírito Santo, Vitória, ES, Brasil

² Centro Brasileiro de Pesquisas Físicas, Rio de Janeiro, RJ, Brasil

³ Centro Federal de Educação Tecnológica Celso Suckow da Fonseca, Itaguaí, RJ, Brasil

⁴ Centro de Pesquisa e Desenvolvimento Leopoldo Américo Miguez de Mello (CENPES), Rio de Janeiro, RJ, Brasil

1 Introduction

Porous rocks are basically heterogeneous systems which are the basic components of oil reservoirs, mineral deposits, soils, and so on [1]. Nuclear magnetic resonance (*NMR*) is a powerful tool commonly used in the study of the geometry and physical–chemical properties associated with the porous space within natural rocks. The technique in essence explores the fact that the surface of the pores and the magnetic properties of the system modify the relaxation rate and the diffusion properties of the fluids within the pores. Studies with *NMR* provide information on the pore-size distributions, surface to volume ratio of the pores, pore-throat distributions, fluid typing, etc. [2, 3]. This information is often deduced from T_2 distributions (describing the relaxation of the transverse magnetization), usually determined through *CPMG* experiments [3, 4]. The characteristic time of the *CPMG* sequence is the inter-echo spacing t_E between two adjacent echoes [3, 4].

The interpretation of *CPMG* results depends on the comprehension of theoretical models describing the physics behind T_2 relaxation experiments. It is well established that for fluids within rock pores, there exist three independent physical mechanisms involved in the M_{xy} relaxation. The bulk relaxation related to fluid interactions (T_{2B}), a surface relaxation that takes into account the interactions of the fluid with the pore surfaces (T_{2S}), and a third one which considers the diffusion process in the presence of magnetic field gradients (T_{2diff}). In principle, these three mechanisms act in parallel leading to the following expression for the relaxation rate $\frac{1}{T_2} = \frac{1}{T_{2B}} + \frac{1}{T_{2S}} + \frac{1}{T_{2diff}}$ [3, 5]. Theoretical models have addressed the physics behind the restricted diffusion in finite size pores, and the free-diffusion limit in the presence of queryIs the word 'ihmomenities' spelled correctly throughout the article? Please check and amend if necessary. magnetic ihmomenities, yielding an expression for the relaxation rate [3, 6–9], namely

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \rho \frac{S}{V} + \frac{D (\gamma G_z t_E)^2}{12}. \quad (1)$$

In Eq. 1, ρ is the surface relaxivity (in $\mu\text{m/s}$), S and V are the surface and volume of the pores, respectively, γ represents the gyromagnetic ratio, while G_z is the magnetic field gradient.

According to the experimental conditions in *CPMG* experiments, three limiting regimes can be used to describe the M_{xy} decay [8–13]. The diffusion length $l_D = \sqrt{D \frac{t_E}{2}}$ measures the typical length that a spin diffuses during an echo interval [9, 10]. The dephasing length $l_G = (\frac{D}{\gamma G})^{1/3}$ corresponds to an average distance over which the spin accumulates a 2π phase shift [9–12], and finally a structural length l_p which can be associated with pore size lengths. Whenever l_D is the lowest among these three characteristic lengths ($l_D \ll (l_G, l_p)$) then the spins can move freely (free diffusion), leading to a magnetization relaxation related to the third term in Eq. 1 [8]. If $l_p \ll (l_G, l_D)$ (small pores) the spins explore the whole porous space many times during a time cycle, averaging out the magnetic field variations, a regime usually identified as motional averaging regime (*MAV*) [8,

11, 14]. On the other hand, when $l_G \ll (l_P, l_D)$ (i.e., for large magnetic field variations) the ensemble of spins loses the phase coherence during diffusion, and this decoherence introduces a significant attenuation of the transverse magnetization. For pores under this condition, the magnetization is not uniform in the whole porous space. This asymptotic behavior is known as localization regime [8–11].

Often pore-size distributions are found through *CPMG* experiments by transforming relaxation time distributions into pore size distributions, assuming that the dominant mechanism is the surface relaxation one. Depending on the experiment, this procedure can be questionable since sedimentary rocks show an intrinsic distribution of internal magnetic fields within the porous space of rocks. This local magnetic inhomogeneity is basically due to the contrast between the magnetic susceptibilities of the fluid and the solid matrix [8, 10], in addition to the possible existence of magnetic impurities.

In this sense, the interpretation of magnetic relaxation due to molecular diffusion in the presence of local non-uniform magnetic fields is a very complicated problem. The principal difficulty in solving this issue is that a realistic description of the variations of the magnetic field within natural rocks is not straightforward as to be described by a simple phenomenological model. Note that two samples can have similar average bulk susceptibilities (i.e., similar phases with equal volume proportions), still showing different gradient distributions due to different spatial distribution of the phases. In other words, local distributions of solid phases within a rock can vary depending of their localization within the rock; therefore, estimating these field distributions for the whole rock by simply observing a local information seems to be a huge challenge. In principle, knowing local fields implies to solve the Maxwell equations at the pore level within the rock by considering spatial dependent susceptibilities [15]. Thus, any approach to this problem can help understand the physics behind the M_{xy} relaxation in the presence of internal magnetic gradients.

Different studies in the last years have focused on the field distribution within natural rocks and its influence on the transverse magnetization relaxation [8–11, 14–16]. In Ref. [15], internal field (and gradient) distributions were simulated based on 2D-section of a sedimentary rock. The gradient distributions were single peak with an average value around zero. These results evidenced the internal field is proportional to the applied external field and the susceptibility difference in the porous media, being bounded inside a field interval. Also an embracing attempt to model magnetic field heterogeneity at the pore space can be observed in [16]. The authors solved Maxwell equations on meshes obtained directly from digital rocks using volume averaged magnetic susceptibilities measured at magnetic fields usually explored in *NMR* experiments. Their results captured some of the principal features observed in magnetic gradient measurements, but the simulations did fail in observing large gradients experimentally measured in rocks. All these results share common features, for example, the local fields are bounded by extreme values which are related to the susceptibilities of the porous media, while at the same time evidence the difficulty of the studied problem. So, as stated in [9], the absence of theoretical models capable of reproducing intrinsic field variations at large scale can be eventually avoided through the use of simulations or heuristic models.

Strong gradients (whatever be the cause) can lead to the emergence of the localization regime (i.e., $l_G \ll (l_p, l_D)$), where the main mechanism governing the relaxation of the magnetization is the strong dephasing between spins. The localization regime occurs at magnetic gradients around tens of miliTesla per meter, which can be easily obtained in scanners used in materials science and other areas [17]. Previous studies suggest this regime is a generic feature of diffusive NMR [9, 17–19], with evidences of a stretched-exponential decay of the overall magnetization for well-defined pore geometries, such as slabs and cylindrical pores [19]. The analysis of this asymptotic regime in natural porous rocks represents a huge challenge due to the complex structure of the pore space in these systems. Thus, theoretical and experimental studies of the localization regime are very important to interpret *NMR* data in the presence of strong magnetic gradients.

In this study, we have simulated the emergence of the localization regime on a real pore structure obtained from X-ray microtomography images of a sedimentary rock. In essence, *CPMG* experiments were simulated on a digital rock containing a distribution of internal magnetic gradients. The novelty of our study lies in the introduction of a real pore space as a suitable geometry to explore the localization regime in *CPMG* experiments. We have also developed a new and simple method that allows the introduction of a local field distribution inside the pore space. This method is new and it incorporates some relevant physics associated with the presence of internal fields inside porous geometries, as reported in the literature (see, for example, reference [15]). Thus, in our simulations, local field distributions can be used as control parameter to identify correlations between local magnetic field gradients (microscopic approach) and macroscopic parameters that can be experimentally determined in *NMR* experiments.

2 Results and Discussion

The digital rock was obtained from μ CT images obtained on a sedimentary rock formation located in Brazil Northeast region, called *Lagoa Salgada*. The used scanner was a GE phoenix vjtoemjx L 300 equipment, and the resolution of the images was $17.2 \mu\text{m}/\text{pixels}$. The images were pre-processed with Matlab techniques and later segmented through a k-means algorithm. This process provided a pore structure (i.e., a digital rock) formed by a 3D matrix containing the solid and pore phases. It is important to stress that the image processing workflow used in this work was aimed to obtain a somewhat realistic pore representation of the rock to be used in our simulations. Thus, we were not concerned with small differences between the obtained pore representation and the real porous geometry within the rock since the unique goal was to afford a realistic pore representation resembling the actual porous structure found within natural rocks. Details regarding the segmentation process used in this work can be found in Ref. [20], while rigorous processing workflows of μ CT images are discussed in [21].

Regarding the digital rock, the resolution of the μ CT images imposes limitations to the size of the pores, i.e., the digital rock does not contain pores with sizes smaller than the μ CT resolution. Although this represents a limitation, the simulations

capture the magnetic relaxation occurring inside large-size pores that are crucial for the flow of fluids since small pores can retain water and other fluids due to their high capillary pressures. In this work, pores were identified with clusters of voxels (i.e., interconnected voxels identified with empty space inside the images) with sizes larger than a cutoff, which was taken as equal to five voxels. This cutoff was used to disregard possible imperfections inside the digital rock arising during the image processing (for example, the existence of isolated voxels identified as pores). It is worth also mentioning that, in our simulations, we have slightly varied the value of the cutoff and the results were not modified. Therefore, our digital rock was formed by large-size pores (i.e., clusters of interconnected empty voxels) including a large cluster that percolates through the whole sample.

Our first objective was the development of a phenomenological method to introduce a realistic intrinsic magnetic field distribution in the context of a computational simulation. Local magnetic field distributions within natural rocks show some general features commonly described in theoretical and experimental approaches to this problem. In fact, we have followed the results reported in [15], where internal field calculations were performed on a Berea sandstone sample by numerically solving local Maxwell equations. In that work, the internal field distribution when scaled to a parameter $\alpha = \frac{B_z - B_0}{\Delta\chi B_0}$ showed a single peak at around $\alpha = 0$, being also bounded in the interval $\alpha \in (-1, 1)$ [15]. In this expression, B_0 is the external applied field, while $\Delta\chi$ represents the difference between the magnetic susceptibilities of the water and the solid matrix. Put differently, this means that the local field is proportional to the applied external field B_0 and also to $\Delta\chi$, being bounded by maximum and minimum fields [15]. Note that the susceptibility of the solid matrix (χ_{ROCK}) should be considered as an average value, possibly containing contributions from different phases.

Following these ideas, in this work the z-component of the internal magnetic induction within the digital rock was randomly introduced (assuming a fully water-saturated rock) by considering a bounded statistical distribution for the dimensionless parameter $\alpha = \frac{B_z - B_0}{\Delta\chi B_0}$. Our ansatz was to simulate α through a beta distribution peaked at $\alpha = 0$, which means $\alpha = 2\alpha_0(\text{Beta}(\beta_1, \beta_2) - 1/2)$. The shape parameters were chosen as $\beta_1 = \beta_2$ (skewness = 0) turning the distribution of the α parameter almost symmetric around $\alpha = 0$. Note that, with this choice, the internal field $B_z = B_0(1 + \alpha\Delta\chi)$ captures the principal features reported in reference [15], in particular, the higher $\Delta\chi$ the higher is B_z . The parameter α_0 determines the bounded limits for B_z , while the parameters $\beta_1 = \beta_2$ control the width of the distribution. It is to be noted that a larger heterogeneity in the composition of a particular rock leads to a wider distribution of internal fields, and this can be incorporated in our model through an appropriated choice of β_1 and β_2 .

To introduce an internal field (z-component) distribution following the method previously described, a tetrahedral mesh was constructed inside the whole digital rock, where a random α value was assigned to each point. These α values were randomly generated from a β distribution with specific parameters ($\alpha_0 = 1$ and β_1). It is important to stress that the choice of $\alpha_0 = 1$ means that the maximum field is bounded by the value $\Delta\chi B_0$ [8, 9]. In the simulations, we took $B_0 = 0.052$

T (currently explored in commercial logging tools) and the water susceptibility as $\chi_{\text{WATER}} = -9.05 \cdot 10^{-6}$ (SI) [22]. The rock susceptibility (χ_{ROCK}) was considered as a free parameter to be chosen according to the conditions of our simulations (for example, rock composition). In essence, this method assigned a field B_z to each point of the mesh. This grid of fields allowed a direct interpolation (using the `interp3` command introduced in MatLab) of the local magnetic field at any position inside the digital rock. Note that, since in this work *NMR* experiments were simulated using a discrete random walk algorithm (of spins), this approach allowed to associate a local field with all spins during their random displacements inside the whole pore space.

In summary, local magnetic fields (quoted by a maximum value) were randomly assigned to the points of a tetrahedral mesh (with resolution lower than the voxel size) inserted in the whole digital rock. From these magnetic fields associated with the mesh points, a local field was determined at any position inside the digital rock. It is to be noted that a more realistic field distribution should consider different maximum gradients for each pore, depending on the pore size. This point is very important and surely a future study in this line would provide important clues regarding the effects of internal fields on *NMR* experiments. Another point that deserves further study concerns the possible existence of intra-pore correlations of the local fields, which are not completely captured by the method used in this work, where random local fields are associated with different mesh points within the voxels; the effects of these correlations on the simulations of the relaxation behavior of the spins within the porous space are to be addressed in future investigations.

According to theoretical models, internal field-strength gradients (G_z) control the diffusion-induced relaxation of the transverse magnetization during *NMR* experiments [3]. Thus, internal magnetic gradients inside the porous space were calculated from the local B_z distributions previously simulated. To this end, random points were positioned within the pore space, and the magnetic gradients (z-direction) were calculated using finite central differences, with spacing equal to the spin displacement used during the random walk. Three examples of G_z distributions inside the porous space can be observed in Fig. 1 (left part). These gradient distributions were computed from B_z distributions that were obtained with the same $\chi_{\text{ROCK}} = 10^{-4}$ [10, 23], but using different $\beta_1 = \beta_2$ parameters, as can be observed in the legend of the figure (right part). It should be stressed that not only the simulated distributions of internal magnetic gradients (G_z) qualitatively resemble other distributions which have been experimentally and/or indirectly determined, as also the calculated gradients are of the same order as those commonly reported in the literature [8, 15, 22, 24]. Thus, in essence, our method allows to introduce a local G_z distribution using an averaged magnetic susceptibility of the solid matrix. Note that in spite of the simplicity of the method, it can be very useful to quantify how a magnetic field heterogeneity can influence the relaxation of the transverse magnetization.

The distributions of internal gradients shown in Fig. 1 can be compared with gradient distributions measured at the same field ($B_0 = 0.05$ T) in sedimentary rocks [9, 16]. The distributions in Fig. 1 are unimodal (reflecting gradients in the bulk of the pores) in agreement with [9] (see Fig. 3a in this reference), while those reported in [16] are multi-modal. It is important to note that, although the model allows the introduction of maximum gradients of the same order of

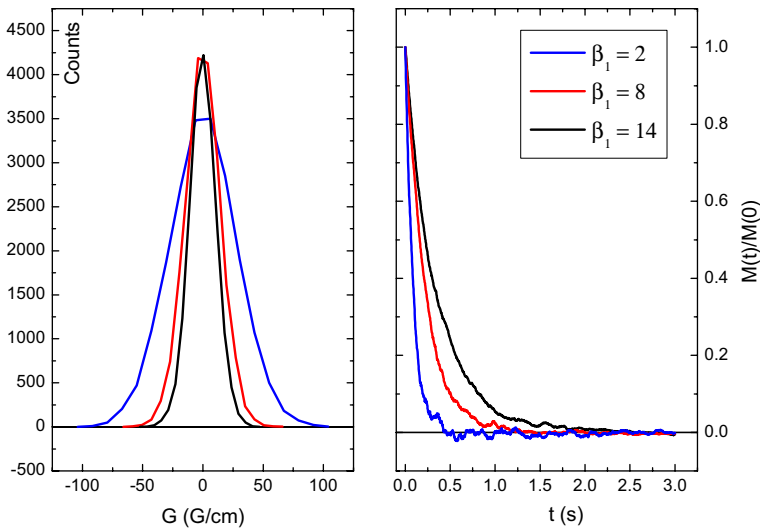


Fig. 1 Left: internal magnetic gradient distributions simulated using the proposed model. Right side: corresponding simulated NMR time-domain signals to each G_z curve (same color) (color figure online)

magnitude (tens of Tesla per meter) as those reported in references [9, 16], in this work, we have assumed a maximum value of 1 T/m. However, a more realistic gradient distribution (yielding larger maximum gradients) would require the inclusion of heterogeneous magnetic susceptibilities, as pointed out in [16]. The comparison indicates that the gradient distributions simulated in this study are in reasonable concordance with experimental results obtained in natural rocks.

The simulations were performed on a digital rock (pore representation) with a volume of approximately 2.6 mm^3 (resolution of $17.2 \text{ }\mu\text{m}/\text{voxel}$). The simulations of the transverse relaxation (M_{xy}) were performed through a discrete random-walk algorithm of diffusive spins as reported in different studies [11, 22, 25–28]. Initially, spins were randomly placed inside the whole pore space formed by voxels forming clusters with sizes above five voxels. This situation simulates a sample fully saturated with water, i.e., a uniform initial spin magnetization. At each time step (taken as $\Delta t = 10 \text{ }\mu\text{s}$), each spin was dislocated randomly a distance $l = \sqrt{6 D_0 \Delta t}$ towards one of their first six neighbors (i.e., along the coordinate axes). In the last expression, $D_0 = 2.5 \cdot 10^{-9} \text{ m}^2/\text{s}$ represented the diffusion coefficient of bulk water. If the new position for a particular spin was inside the solid matrix (i.e., if the spin found a pore-solid interface), then the walker (spin) was killed with probability $P = \frac{\rho l}{A D_0}$, where ρ is the surface relaxivity (in $\mu\text{m}/\text{s}$) and $A \approx 1$ [29], otherwise the walker returned to its last position. At each time step (Δt), each spin accumulated a phase difference $\Delta\Phi = \gamma \Delta t (B_z - B_0)$, where γ represented the gyromagnetic ratio for protons and B_z was the interpolated magnetic field experimented by each spin at its local position. Periodic boundary conditions were used on the spin ensemble during the random walk to simulate bulk

conditions. Our simulation attempted to reproduce a *CPMG* experimental sequence assuming a characteristic time between π pulses $t_E = 2\tau$, or equivalently a time 2τ between consecutive echoes. This experimental sequence was simulated by inverting the phases of all spins during the simulation whenever the elapsed time satisfied the condition $t = n\tau$ with n odd, while for n even an echo signal was recorded with intensity given by

$$\frac{M(t = n\tau)}{M(0)} = \exp\left(-\frac{n\tau}{T_{2B}}\right) \frac{1}{N(n\tau)} \sum_{i=1}^{N(n\tau)} \cos\{\Delta\Phi_i(n\tau)\}. \quad (2)$$

In the last equation, $M(0)$ is the initial magnetization (initial number of walkers); T_{2B} ($= 3.5$ s) represents the spin-spin relaxation time of the pore fluid (bulk relaxation); $N(n\tau)$ is the number of surviving walkers (spins) at time $t = n\tau$ and $\Delta\Phi_i(n\tau)$ describes the phase accumulated by the spin i at time $t = n\tau$. Note that in essence, our results basically depend on the surface relaxivity (ρ) and the averaged magnetic susceptibility (χ_{ROCK}), both being intrinsic properties of the rock. The results also depend on the characteristic time of the experiment (t_E).

On the right side of Fig. 1 are shown some *NMR* simulations (obtained by the method described above) using the corresponding gradient distributions appearing on the left side of the figure. In these simulations, $t_E = 210^{-3}$ s, while the G_z distributions were introduced by taking $\chi_{ROCK} = 10^{-4}$ and $\chi_{WATER} = -9.05 \cdot 10^{-6}$ [10, 22, 23]. Also, the relaxivity was taken as $\rho = 30 \mu\text{m/s}$ [30]. At first glance, it is clear that a direct correlation exists between the relaxation rate and the heterogeneity of the distributions, i.e., the broader is the G_z distribution (lower β_1), the faster is the M_{xy} relaxation. This is an expected result since in theory the diffusion-induced relaxation rate ($1/T_{2 \text{ diff}}$) should be directly proportional to the square of the strength of the magnetic field gradient, i.e., $\frac{1}{T_{2 \text{ diff}}} = \frac{D (\gamma G_z t_E)^2}{12}$ [3].

Another interesting point regarding *NMR* experiments is that molecular diffusion in a rock occurs inside pores and also between pores (interpore coupling). Usually, theoretical considerations are developed by considering the weak coupling regime, where the interpore diffusion is neglected while the overall decay in the transverse magnetization is controlled by relaxation processes simultaneously occurring in isolated pores with different sizes. Even if this scenario might not be strictly applicable to natural rocks, it has been often used to extract information in *NMR* experiments [31]. In our study, the segmentation of the μCT images provided a digital rock containing an ensemble of isolated pores (weak coupling regime). Each isolated pore was formed by a particular group of interconnected voxels, where each voxel shares one face with at least another voxel of the same pore. Thus, our *NMR* simulations reproduced the relaxation of the magnetization under the weak coupling regime.

In this work, we also have performed *NMR* simulations considering different inter-echo spacings (i.e., different t_E). The simulations were done using a constant relaxivity parameter $\rho = 30 \mu\text{m/s}$ and also $\chi_{ROCK} = 10^{-4}$. The porous space was considered as formed by clusters (i.e., pores) with sizes above a critical threshold of five voxels. A field distribution was introduced within the digital rock by using parameters $\beta_1 = \beta_2 = 8$ and $\alpha_0 = 1$. It should be noted that this microscopic field

distribution (and its corresponding G_z distribution) was kept unaltered during all simulations at the different t_E values. This reproduces an experimental situation in which a fully saturated rock sample is sequentially analyzed under similar conditions, but at different inter-echo spacings.

In Fig. 2 are shown some of the the $M_{xy}(t)$ relaxation decay signals and their corresponding T_2 distributions calculated for each $M_{xy}(t)$ curve (same color). The inter-echo spacing was the unique modified parameter, with values ranging from hundreds of μs up to ms . The T_2 distributions were calculated by implementing a Tikhonov regularization algorithm as described in [32]. In this algorithm, a penalty function is imposed on the recovered solution aiming to stabilize the numerical inversion problem used to obtain the T_2 distribution. This penalty correction is controlled by a regularization parameter λ . The optimal value of λ was automatically found using the L-curve method. In this approach, the optimal λ is located close to the corner (i.e., maximum curvature) of the curve that is obtained from the parametric plot (log–log plot) of the residual norm against the solution norm. Details on the method can be found in reference [32] and references therein. It can be observed in the plots shown in Fig. 2 that the T_2 distributions are shifted to lower values as t_E is increased. In principle, this indicates that longer echo spacings in the simulations allow a greater influence on the spin ensemble of the magnetic inhomogeneities (diffusive attenuation) and/or pore surfaces, causing a faster relaxation of M_{xy} .

To shed some light on how the local gradient distribution determines the dynamics of the spins ensemble we have characterized each T_2 distribution by its logarithmic mean, defined as $T_{2LM} = 10^{\frac{\sum A_0 \log(T_2)}{\sum A_0}}$, where A_0 represents the amplitudes observed in Fig. 2 for each T_2 value. The relaxation rate defined as $\frac{1}{T_{2LM}}$ is plotted as a

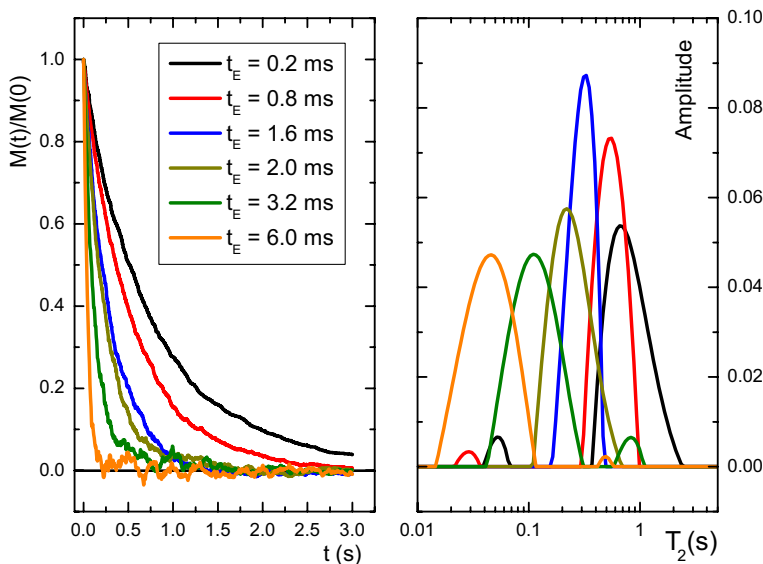


Fig. 2 NMR simulations for different inter-echo spacings (left) and their correspondent T_2 distributions (right side)

function of t_E^2 in Fig. 3. The graph includes a linear fit ($R^2 = 0.992$) of the data for low t_E^2 values, while the inset shows a semi log plot of the same data. Qualitatively, these results evidence the existence of a linear behavior of T_{2LM} up to a certain critical value t_{Ec}^2 from which the behavior changes to some type of logarithmic dependence.

To obtain information from Fig. 3, it is necessary to explore the different relaxation regimes usually discussed in the literature. It is well established that for fluids occupying the porous space of rocks there exist three independent physical mechanisms determining the relaxation of the transverse magnetization. The bulk relaxation quantifies bulk fluid interactions (T_{2B}); a surface relaxation takes into account the interactions of the fluid with the surface of the pores (T_{2S}); and a third one that considers the diffusion process in the presence of magnetic field gradients (T_{2diff}) [3]. Theoretical models have addressed the physics behind the restricted diffusion in finite-size pores, and the free-diffusion limit in the presence of magnetic inhomogeneities, yielding Eq. 1 for the relaxation rate [3].

The main mechanism that determines the M_{xy} relaxation depends on the shortest characteristic length associated with the different processes in Eq. 1. Considering spherical shape pores, and using those with smaller sizes (i.e., 5 voxels and a resolution of $17.2 \mu\text{m}/\text{pixel}$), we obtain $l_S \approx 18.2 \mu\text{m}$. In turn, since $l_G \sim \frac{1}{G_z^{1/3}}$ then using the largest gradient of the distribution used in our simulations $G_{zmax} \approx 0.6 \text{ T/m}$ (see Fig. 1), we can estimate $l_{Gmin} = (\frac{D_0}{\gamma G_{zmax}})^{1/3} \approx 2.4 \mu\text{m}$. On the other hand, the diffusion length $l_D = \sqrt{D \frac{t_E}{2}}$ depends on t_E , and it quantifies an average distance that a spin diffuses during the echo time. Thus, according to our conditions, whenever the

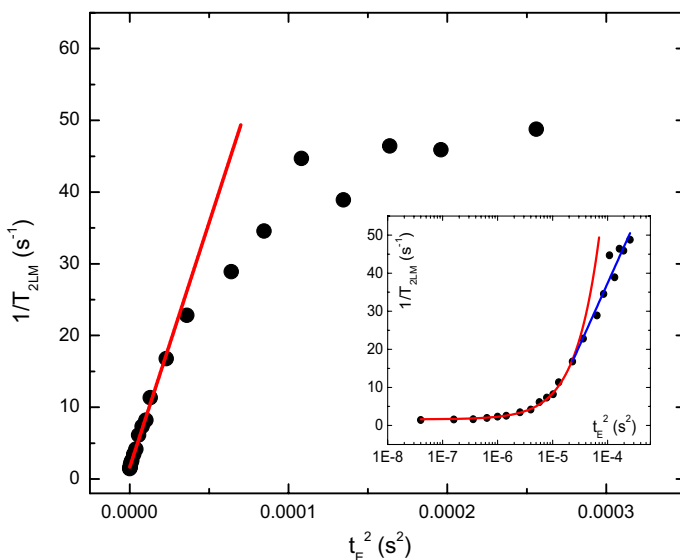


Fig. 3 Main plot: relaxation rate ($1/T_{2LM}$) as a function of t_E^2 . Inset: same data in semi-log plot. Straight lines (red and blue lines) represent linear fits of the data in different regions (for low and high t_E values, respectively) (color figure online)

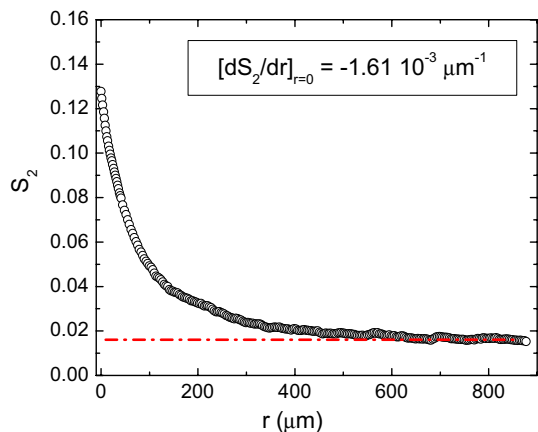
condition $l_D < l_G$ is met, then the NMR simulations reproduce well the relaxation processes mainly occurring in the free-diffusion limit. In our simulations, as t_E increases l_D also increases approaching l_{Gmin} . Thus, we can define a critical inter-echo spacing (identified as t_{Ec}) at which the condition $l_D(t_{Ec}) = l_{Gmin}$ is verified. This condition allows finding the critical threshold $t_{Ec} = \frac{2 l_{Gmin}^2}{D} \simeq 5.0 \cdot 10^{-3} \text{ s}$. Below t_{Ec} the results of the simulations can be analyzed under the frame of the free-diffusion model, while above it, localization effects should influence the results.

In Fig. 3, the linear behavior of the data can be observed for $t_E \leq 4.8 \cdot 10^{-3} \text{ s}$ ($t_E^2 \leq 2.3 \cdot 10^{-5} \text{ s}^2$), which is just around t_{Ec} . Thus, the linear behavior in this low- t_E region can be analyzed considering the free-diffusion regime, i.e., the third term in Eq. 1. The red line in Fig. 3 represents a linear fit of the data for $t_E \leq t_{Ec}$. This fit provided the linear and angular coefficients $a = (1.6 \pm 0.2) \text{ s}^{-1}$ and $b = (6.8 \pm 0.2) 10^5 \text{ s}^{-3}$, respectively.

Taking into account the expression in Eq. 1 and the angular coefficient $b = 6.8 \cdot 10^5 \text{ s}^{-3}$, in addition to $D_0 = 2.2969 \cdot 10^{-9} \text{ m}^2/\text{s}$, and the gyromagnetic ratio for protons $\gamma = 2.67 \cdot 10^8 \text{ rad T}^{-1} \text{ s}^{-1}$ (as used in our simulations), we have estimated an average magnetic gradient $\langle G_z \rangle = 0.22 \text{ T/m}$, which is a reasonable effective gradient according to the field distribution used in the simulation (see Fig. 1, with $\beta = 8$).

On the other hand, by considering the linear coefficient $a = 1.6 \text{ s}^{-1}$; $\rho = 30 \text{ } \mu\text{m/s}$ and $T_{2B} = 3.5 \text{ s}$ (as used in our simulations), we obtain $(S/V)_p = 4.4 \cdot 10^{-2} \text{ } \mu\text{m}^{-1}$. The parameter $(S/V)_p$ represents the surface to volume ratio of the pores. To verify these results, the two-point correlation function (or autocorrelation function) was calculated from the digital rock model used in our simulations [33], as shown in Fig. 4. The two-point autocorrelation function for the phase pore, identified as $S_2(r)$, represents the probability that two random points separate by a distance r be pores. If the porous medium is statistically homogeneous and isotropic (like the rock explored here), $S_2(r)$ does not depend on the relative orientation of the vector connecting both points, and it is invariant under a translation operation [33]. One interesting property of this statistical descriptor is given by $\left\{ \frac{d S_2(r)}{dr} \right\}_{r=0} = -\frac{s}{4}$, where $s = \frac{S_p}{V_T}$ represents the surface to volume of the pores. Thus, it is possible to use the $S_2(r)$ function to find $(S/V)_p$ by

Fig. 4 Two-point autocorrelation function computed on the digital rock. The red line marks the squared of the porosity (color figure online)



rewriting $\left\{\frac{dS_2(r)}{dr}\right\}_{r=0} = -\frac{1}{4} \frac{S_P}{V_P} \frac{V_P}{V_T}$, where $\phi = \frac{V_P}{V_T}$ is the porosity of our digital rock. Therefore, ten $S_2(r)$ curves were computed and an average for the first derivative in the limit $r \rightarrow 0$ (linear behavior) was calculated. This value was $\left\{\frac{dS_2(r)}{dr}\right\}_{r=0} = -1.6 \cdot 10^{-3} \mu\text{m}^{-1}$, from which it was obtained $(S/V)_P = 5.05 \cdot 10^{-2} \mu\text{m}^{-1}$, a value similar to the one obtained from the analysis of Fig. 3.

Regarding the high- t_E regime shown in Fig. 3, it is to be stressed that as t_E increases l_D turns to be larger than l_G , but still significantly lower than l_S . This condition indicates that a new regime is dominating the M_{xy} decay, specifically the localization limit (non-homogeneous magnetization in the pores). Nevertheless, in this asymptotic limit theoretical models indicate T_2 does not depend on the inter-echo spacing, which does not agree with the results shown in Fig. 3. Thus, we suggest that the second linear regime observed in Fig. 3 for higher t_E values corresponds to a transition between two asymptotic limits. This idea is not new; in fact, it has been previously suggested, where the existence of a preasymptotic localization regime with irrelevant surface relaxation could lead to some dependence of the relaxation rate on the inter-echo spacing [9]. As stated in the same work [9], no theoretical model has been developed to describe this physical situation.

Thus, we assume that the non-linear behavior observed in Fig. 3 for $t_E \geq t_{Ec}$ can be associated with the non-homogeneity of bounded magnetic fields within the sample. This means that whichever the position within the porous space, there exists a different local gradient affecting the spins that diffuse in this region. From a physical viewpoint, this idea means that the temporal dependence of M_{xy} should be controlled by the local details of the magnetic structure of the rock (i.e., averaged local gradients). Therefore, at t_{Ec} spins at local regions within the pores (away from the pore walls) that experience the maximum effective gradient of the distribution completely dephase, while the remaining spins proceed to relax under the free-diffusion limit. In other words, the signal is formed by isolated pockets of surviving magnetization [18]. As t_E increases, a higher proportion of spins diffuse in regions where they loss completely their coherence (i.e., $l_G < l_D$), while other spins remain in pockets with small effective gradients. These spin bundles contribute to the dependence between the relaxation rate and t_E^2 observed in Fig. 3.

Any quantitative analysis for the high- t_E data in Fig. 4 requires, in principle, an analytical model for the M_{xy} decay in the preasymptotic localization regime. In this sense, the lack in the literature of theoretical models in this region turns difficult any quantitative conclusion concerning the observed behavior. Nevertheless, the data in the inset of Fig. 3 suggest, at first sight, a logarithmic dependence between the relaxation rate and the inter-echo spacing. Based on this we can propose an heuristic model for this transition region according to

$$\frac{1}{T_2} = \frac{\alpha_1}{t_{Ec}} \ln \left\{ 1 + \alpha_2 \left(\frac{t_E}{t_{Ec}} \right)^2 \right\}. \quad (3)$$

In Eq. 3, the non-dimensional constants $\alpha_1 = \frac{G_z}{G_{zmax}} < 1$ and α_2 are free parameters of the equation. Note that for $t_E \ll t_{Ec}$ we have $\ln \{ 1 + \alpha_2 (\frac{t_E}{t_{Ec}})^2 \} \approx \alpha_2 (\frac{t_E}{t_{Ec}})^2$, from which the classic expression $\frac{1}{T_2} \propto D \gamma^2 G_z^2 t_E^2$, valid in the free-diffusion limit, is recovered.

In Fig 5, we show the same data previously shown in Fig. 3, but including also a fit with the proposed model (Eq. 3). As it can be observed there exists a reasonable match between the data and the model, at least for the explored details in our simulations. Still, we stress that this model in principle does not indicate or points to any particular physical mechanism as the responsible for the transverse magnetization relaxation, and further work (including both experiments and simulations) is necessary to clarify our findings in this transition region.

It should be highlighted that the analysis of NMR data in the framework of the localization regime can be crucial to determine accurate petrophysical properties of reservoir rocks [18]. Our simulations clearly indicate that T_2 distributions can be significantly shifted towards lower values, depending on the existence of appreciable magnetic field gradients inside the rocks. Thus, in the case of rocks with expressive magnetic susceptibilities, parameters of the pore structure commonly determined from T_2 distributions (such as pore-size distributions and surface to volume ratios, for example) can be misinterpreted if these gradients are not properly considered. This can induce to wrong results for the estimation of macroscopic parameters, for example, porosity or permeability [18]. Concerning this point, an appropriate analysis of T_2 distributions, considering the physics behind the localization regime, can shed some light on the magnetic structure of porous rocks. In this sense, the analysis of statistical features of T_2 distributions (logarithmic mean, modal value, etc.), in addition to other experimental parameters (for example, t_{Ec} in Fig. 5) can help to quantify internal gradients and consequently magnetic susceptibilities. Ultimately, this can be an indicative of the possible existence of magnetic phases inside the rocks, which can be very useful to classify natural rocks.

Finally, it would be interesting to investigate specific aspects related to the transition between the pre-asymptotic and localization regimes. In this line, the

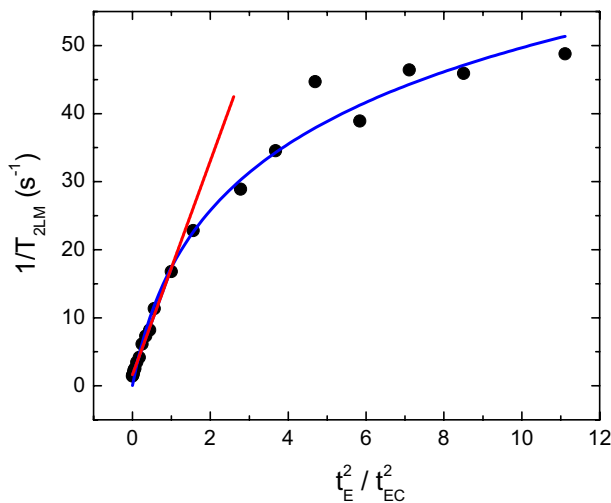


Fig. 5 Same data in Fig. 3 plotted as a function of $\frac{t_E^2}{t_{Ec}^2}$. The blue curve is a fit with Eq. 3 and parameters: $t_{Ec} = 4.8 \cdot 10^{-3}$ s; $\alpha_1 = 0.08$; $\alpha_2 = 1.81$ (color figure online)

simulations could be reproduced on simple porous geometries (spherical pores, for example) to separate the influence of features associated with the reconstruction of the digital rock.

3 Conclusions

In summary, we have simulated the transverse magnetization relaxation in a real porous structure obtained from X-ray microtomography images of a sedimentary rock. The simulations included a local distribution of magnetic fields within the porous geometry. The field distribution was simulated through a statistical approach, which introduced some properties commonly reported in other studies related to the presence of magnetic non-homogeneities in natural rocks. Still, the simulations were conducted under particular conditions assuming that the relaxation of the transverse magnetization was mainly associated with diffusion processes occurring in the presence of strong magnetic gradients. Our results show a striking influence of the magnetic gradient distribution on the relaxation rate (characterized by the logarithmic mean of the T_2 distribution). The relaxation rate depends linearly on the square of the inter-echo spacing at low values (free-diffusion regime), while this dependence turns to be somewhat logarithmic for high t_E values. This high- t_E regime was associated with a pre-asymptotic localization regime previously suggested in the literature. Lastly, a phenomenological model was used to fit the data in all range of t_E values simulated in this work. Our work points out to the importance of exploring this pre-asymptotic relaxation regime, which can be experimentally investigated in future work.

Acknowledgements This work was made possible by cooperation agreement between CENPES PETROBRAS and CBPF (cooperation agreement 5850.0106789.18.9).

References

1. M. Sahimi, *Rev. Mod. Phys.* **65**, 1393 (1993)
2. Y.Q. Song, *N. J. Phys.* **14**, 055017 (2012)
3. G.R. Coates, L. Xiao, M.G. Prammer, *NMR Logging Principles and Applications* (Halliburton Energy Services, Gulf Publishing Company, Houston, 1999)
4. H.Y. Carr, E.M. Purcell, *Phys. Rev.* **94**, 630 (1954)
5. W.E. Kenyon, *Log Anal.* **38**, 21 (1997)
6. K.R. Brownstein, C.E. Tarr, *Phys. Rev. A* **19**, 2446 (1979)
7. P. Bendel, *J. Magn. Reson.* **86**, 509 (1990)
8. M.D. Hurlimann, *J. Magn. Reson.* **131**, 232 (1998)
9. J. Mitchell, T.C. Chandrasekera, M.L. Johns, L.F. Gladden, E.J. Fordham, *Phys. Rev. E* **81**, 026101 (2010)
10. D. Grombacher, E. Fay, M. Nordin, R. Knight, *Geophysics* **81**, EN43 (2016)
11. R.M.E. Valckenborg, H.P. Huinink, J.J. van der Sande, K. Kopinga, *Phys. Rev. E* **65**, 021306 (2002)
12. M.D. Hurlimann, K.G. Helmer, T.M. de Swiet, P.N. Sen, C.H. Sotak, *J. Magn. Reson.* **113**, 260 (1995)
13. P.R. Connolly, S.J. Vogt, S. Iglauer, E.F. May, M.L. Johns, *Water Resour. Res.* **53**, 7917 (2017)
14. L.J. Zielinski, P.N. Sen, *J. Magn. Reson.* **147**, 95 (2000)
15. Q. Chen, A.E. Marble, B.G. Colpitts, B.J. Balcom, *J. Magn. Reson.* **175**, 300 (2005)

16. P.R.J. Connolly, W. Yan, D. Zhang, M. Mahmoud, M. I Verrall, M. Lebedev, S. Iglauer, P.J. Metaxas, E.F. May, M.L. Johns, J. Petrol. Sci. Eng. **175**, 985 (2019)
17. R. Valiullin, ed., *Diffusion NMR of Confined Systems. Fluid Transport in Porous Solids and Heterogeneous Materials* (RSC Publishing, Cambridge, 2016)
18. E.J. Fordham, J. Mitchell, Microporous Mesoporous Mater. **269**, 35 (2018)
19. N. Moutal, K. Demberg, D. Grebenkov, T.A. Kuder, J. Magn. Reson. **305**, 162–174 (2019)
20. J.L. Gonzalez, E.L. de Faria, M.P. Albuquerque, M.P. Albuquerque, C.R. de Bom, J.C.C. Freitas, C.W. Cremasco, M.D. Correia, J. Petrol. Sci. Eng. **166**, 906 (2018)
21. S. Schluter, A. Sheppard, K. Brown, D. Wildenschild, Water Resour. Res. **50**, 3615 (2014)
22. C.H. Arns, T. AlGhamdi, J.-Y. Arns, N. J. Phys. **13**, 015004 (2011)
23. D.A. Clark, D.W. Emerson, Explor. Geophys. **22**, 547 (1991)
24. B. Sun, K.J. Dunn, Phys. Rev. E **65**, 051309 (2002)
25. K.S. Mendelson, Phys. Rev. B **41**, 562 (1990)
26. M. Leibig, J. Phys. A **26**, 3349 (1993)
27. J. Finjord, A. Hiorth, H. Unn, S.M. Skjæveland, Transp. Porous Med. **69**, 33 (2007)
28. J. Guo, R. Xie, Y. Zou, Y. Ding, J. Geophys. Eng. **13**, 285 (2016)
29. D.J. Bergman, K.J. Dunn, L.M. Schwartz, P.P. Mitra, Phys. Rev. E **51**, 3393 (1995)
30. E. Lucas-Oliveira, A.G. Araujo-Ferreira, W.A. Trevizan, C.A. Fortulan, T.J. Bonagamba, J. Magn. Reson. **292**, 16 (2018)
31. C.H. Arns, A.P. Sheppard, R.M. Sok, M.A. Knackstedt, Petrophysics **48**, 202 (2007)
32. I.J. Day, J. Magn. Reson. **211**, 178 (2011)
33. S. Torquato, Annu. Rev. Mater. Res. **32**, 77 (2002)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.